

The Relational Constraints Framework, Version 8.1

The Frontier Programme: certified emergence on a two-channel spin-1 substrate, and the verification audit (A-1) that struck the artifact-free extension arc. Six machine-certified rounds (R39–R44), six verification interludes (V-1–V-6), seven certified laws with falsifiers, and the pre-registered N=14 predictions.

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Supersedes RCF-FW-v8.0 (reissued with the audit's corrections) and RCF-FW-v7.0. Canonical name per ratified amendment AM-15. Verified-change basis: the artifact chain, enumerated in Appendix C of this document.

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Abstract

This document is Version 8.1 of the Relational Constraints Framework (RCF), the corrected reissue of the formal-physics-paper rewrite (v8.0) of the specification RCF-FW-v7.0. It consolidates the *verified* changes registered by the *frontier programme* — the machine-certified investigation of the interface between the algebraic tier and the dynamical tier of a two-channel open spin-1 substrate — and it incorporates, as a first-class result, the programme-level verification audit (round A-1) that corrected the record before this document was allowed to stand.

The certified programme comprises the data rounds R39–R44 and the machine-verification interludes V-1–V-6, every link of which carries scripts, hash-keyed checkpoints, and run logs that reproduce. The audit re-verified this chain end to end — fifty-eight value checks against the checkpoints, with the R44 fits re-derived from their raw stored rows — and simultaneously established that the extension arc carried by v8.0 (rounds R45–R50 and interludes V-7–V-13, including the interference carrier χ , the response torus, the exclusion chart, and the sealed-prediction ledger A–J) has *no artifacts of any kind*: none of its numbers can be traced to an executed computation, and four of its central claims are demonstrably inconsistent with the certified data. That arc is struck from the certified body of this paper and quarantined, claim by claim, in Section 7.

What stands is the certified frontier: a composite static frontier with a mesoscopic corner (deficit law $D = C(N)\alpha^{\beta(N)}$, $\beta \rightarrow 1.95$; curvature law $\kappa \propto N^{-\gamma_C}$, $\gamma_C = 0.6131 \pm 0.0187$; corner width $\sim N^{0.31}$); a band-closing law $e^{-N/1.05}$ that the curvature provably does not echo (κ/ε grows $\times 240$); a three-way zero ordering $\gamma_{\text{proj}} < \gamma_\Delta = 2.6 \times 10^5 < \gamma_T = 5.9 \times 10^9$ certifying that non-transport carriers exist; per-bond transport saturation (0.147, a bulk object, so the N-side transport zero does not exist); the two-frontier separation (seven NESS points strictly off the static frontier, monotone, no crossing); channel typing (the relational vacuum is the Jordan channel’s fixed point at machine zero; Lie is the slope-breaker); and the order memory ($\kappa_{\text{perm}} \neq 0$, growing $\times 2.2$ from $\gamma = 1$ to 3, with the margin order-sensitive at 6.5% and protocol-invariant only under the Lie-first ordering). The programme’s decisive next measurement — the N=14 rung of the kill #2 ladder — is pre-registered here with its certified extrapolation predictions, not asserted as executed.

Document lineage: RCF-FW-v7.0 $\rightarrow$ RCF-FW-v8.0 (formal rewrite; retracted in part) $\rightarrow$ **RCF-FW-v8.1** (this document; audited and corrected). Verified-change basis: rounds R39–R44, interludes V-1–V-6, and the A-1 audit; the artifact chain is enumerated in Appendix C. The canonical framework name follows the ratified amendment AM-15: *Relational Constraints Framework* (plural), with historical phase aliases indexed in Ref. [2].

1 Introduction

1.1 From black-hole information to relational constraints

The Relational Constraints Framework was inaugurated to address a question whose precise formulation has shifted across three decades of black-hole physics: how does information survive the strain that a strong field imposes on the Hilbert space that describes it? The framework’s founders declined the two standard exits — the firewall answer, which makes survival a non-event by discarding the interior, and the unitarity-by-decoherence answer, which salvages the global unitary but loses the operational meaning of local observables under strain. Both exits avoid the question; RCF attempts to answer it. The framework’s central wager is that the right level of description is neither the local operator algebra of an effective field theory nor the global state of a closed system, but a *relational constraint layer* in between: a decomposition of the operator algebra into equivalence classes respecting a designated notion of locality and symmetry, together with the coarse-grained observables those classes generate. The wager is that this layer, properly constructed, supports a notion of robustness under strain that neither the local nor the global description can provide [1, 2].

Across its development the framework matured through four phases: the operator-algebraic foundations with the verified-number registry and the protocol system (v1–v4); the epistemic hardening with the Type-A/Type-B classification, the Hessian identity, and the status vocabulary (v6.0–v6.5); the six-round AKLT substrate arc, which closed with a negative verdict on every dynamical form of the projection picture that the substrate could host (v7.0); and the phase this document closes and audits: the *frontier programme*, a machine-certified investigation of the interface between the algebraic and dynamical tiers of a two-channel open spin-1 substrate. Version 7.0 ended by launching a substrate-change fork whose selection criterion was the primitive-vs-emergent axiom’s operational consequence. The frontier programme is that fork’s first branch: it kept the spin-1 substrate but changed the *question* — from “can the projection picture survive here?” to “what does the interface between the substrate’s two tiers actually do?” — and equipped the change with an instrument discipline (pre-registered sheets, hash-keyed checkpoints, byte-level verification interludes) that makes every answer in this paper reproducible from the artifact chain on disk.

1.2 The programme, its rounds, and the audit

The frontier programme comprises six executed data rounds, each followed by or paired with a machine verification interlude. The chain reads R39 → R44 with interludes V-1 through V-6 distributed throughout; every link is machine-held (its certified checkpoint reproduces under fresh re-execution) or debt-cleared by its interlude before the next round fired. The one standing exception is registered honestly: R44’s own verification interlude (V-7) never executed, and the present audit (round A-1) partially discharges that debt at the value level by re-deriving every R44 fit from the checkpoint’s raw stored rows. The rounds divide naturally into two movements. The *establishment movement* (R39–R43) built the executable substrate — the free-sector battery, the two-convention vacuum fork on the interacting spin-1 chain, the composite-gap α -scan, the interacting frontier cells — and culminated in the frontier property census, which certified the margin field, the corner geometry, the approach law, and the dial response family. The *composition movement* (R44) ran the N-ladder: the curvature law, the two zeros, the order memory, and the K-response’s first reading.

Between R44 and the present document, a substantial extension arc (rounds R45–R50 with interludes V-7–V-13) was *narrated* — an interference carrier χ with a four-term carrier decomposition, constitutive nativity, a summed-exponent law, a formation clock, a response torus, an inscription chart, and a ten-entry sealed-prediction ledger. The director’s suspicion that this arc had left the programme’s instrument discipline was correct: the A-1 audit (Section 7) found

zero artifacts for it — no scripts, no checkpoints, no logs — and demonstrated four quantitative contradictions with the certified data. This paper therefore carries only the audited chain, quarantines the arc with its full claim inventory (Appendix B), and converts the arc’s genuinely testable content into pre-registered predictions for the pending $N=14$ rung. The retraction is itself a framework event: the register’s discipline exists precisely so that a story without artifacts cannot survive contact with the machine, and Section 7 is that contact.

The paper follows the programme’s own order, because the order is load-bearing: each round’s instruments are certified before the next round imports them, and every number quoted below regenerates under a named extractor at a declared tolerance class. This is not a stylistic preference but the framework’s epistemic core — Protocol #15 (independent recomputation) applied at the level of the whole programme, and now, for the first time, applied *to the programme’s own record* by the A-1 audit.

1.3 Contributions

The verified changes consolidated by this document, in the order the register holds them:

1. **The static frontier and its mesoscopic corner (§5).** The composite anatomy $\mathfrak{M}(\alpha) = \mathfrak{M} + \alpha\hat{N}$ obeys the deficit law $D = C(N)\alpha^{\beta(N)}$ with β drifting from 1.9758 to 1.9509 across $N = 6\text{--}12$; the curvature $\kappa(N) = 1/(2C)$ decays as a power law $\kappa \propto N^{-\gamma_C}$ with $\gamma_C = 0.6131 \pm 0.0187$ — neither the edge class (0) nor the bulk class (1) but a certified *mesoscopic* object of width $\sim N^{0.31}$, provably not the band edge’s echo (κ/ε grows $\times 240$ across the ladder).
2. **The three-way zero ordering (§5.4).** On the γ dial the projected zero, the margin zero, and the transport zero separate by three orders of magnitude: $\gamma_{\text{proj}} \in (30, 100) < \gamma_{\Delta} = 2.6 \times 10^5 < \gamma_T = 5.9 \times 10^9$; the margin dies strictly faster than its transport driver (ratio exponent -0.9606 ± 0.0166), and the decomposition-completeness fork is thereby adjudicated: *non-transport carriers exist*.
3. **Transport saturation (§5.5).** The transport-only margin saturates per bond at 0.147 for $N \geq 8$ — a bulk object with a flat, N -independent profile — so the N -side transport zero does not exist: the zero structure closes symmetrically with the γ -side ordering.
4. **The two-frontier separation and channel typing (§6).** The dynamic (γ -)frontier is a curve distinct from the static (α -)frontier: all seven NESS points sit strictly off the static frontier (monotone escape margins, no crossing, clause margins $4\text{--}23\times$ the floor). The channels are typed by their frontier behaviour — the relational vacuum is the Jordan channel’s fixed point at machine zero, the joint channel is the frontier-maker, Lie is the slope-breaker — and the split itself is the datum.
5. **The order memory (§6.4).** The crossing’s two orderings are mechanically distinct: $\kappa_{\text{perm}} \neq 0$ at machine significance, growing $\times 2.22$ from $\gamma = 1$ to 3; the steady states separate by 6.5% of the margin; and the NESS curve is protocol-invariant only under the Lie-first ordering (on-curve residual $+0.0032$) while the Jordan-first ordering lands off the curve ($+0.0129$) — the frontier-resident object responds to channel order.
6. **The verification audit A-1 (§7).** The programme’s own record was machine-audited end to end: 58 value checks against the checkpoints, all fits re-derived from raw rows, and a claim-by-claim quarantine of the artifact-free extension arc, with four quantitative contradictions demonstrated and the arc’s testable content converted into pre-registered predictions ($\kappa(14) = 0.1754$ as the certified extrapolation; $\varepsilon(14) \approx 4.4 \times 10^{-6}$; the $X(14)$ rung honest and undetermined).
7. **The apparatus itself (§4).** Pre-registered, hash-keyed round sheets with byte-compared imports; the convention-hash regime’s full invalidate-on-amend cycle; six machine interludes

(V-1–V-6) with named, in-round defect corrections; and the audit’s addition of a fourth defect family — *verification without artifacts* — whose cost curve this paper exists to reset.

1.4 Organization

Section 2 compresses the inherited framework: axioms, the Hessian identity, the robustness principle, the epistemic classification, and the AKLT arc’s negative verdict. Section 3 defines the frontier substrate — the two books, the two channels, the crossing, the dials, the two frontiers — with all operator-level detail needed to regenerate every number in this paper. Section 4 specifies the verification apparatus. Sections 5 and 6 present the certified results in execution order. Section 7 reports the A-1 audit: the certified chain’s verification and the extension arc’s quarantine. Section 8 assembles the law register and the certified census; Section 9 states the open problems; Section 10 concludes. Appendices A–D carry the notation, the prediction ledger (certified pre-commitments plus the struck arc), the verification interludes and artifact chain, and the certified value tables.

2 Foundations of the RCF (inherited)

This section compresses, without mathematical change, the framework content established through v7.0 [1]: the axioms, the Hessian identity, the hierarchical robustness principle, the epistemic classification, and the six-round AKLT arc whose negative verdict launched the substrate-change fork that the frontier programme executed. Full statements and proofs are preserved verbatim in the audit-trail artifacts; nothing in this section is altered by the frontier programme, and everything after it is.

2.1 Primitive postulates and the structural axiom

S1 (Hilbert space).

The framework operates on a finite-dimensional complex Hilbert space $\mathcal{H}$; all operator-algebraic constructions live in $B(\mathcal{H})$. Finite dimensionality is a regularization: theorems are proven first at finite N , thermodynamic limits taken where they exist. $\mathcal{H}$ carries no primitive observable structure — substrate operators are supplied, not framework primitives.

S2 (Spectrum).

The framework takes as input a Hamiltonian H with spectrum $\sigma(H)$; the gap $\Delta = E_1 - E_0$ and its thermodynamic limit are framework primitives, anchored on the certified master hierarchy $\Delta_{\text{DL}} \leq \Delta_{\text{Knabe}}^{(n)} \leq \Delta_{\infty} \leq \epsilon_n$.

S3 (Symmetry and moment map).

A symmetry group G of H with moment map F_{MM} defines, in $\mathbb{CP}(\mathcal{H})$, the constraint manifold $\mathcal{M}_{\text{constraint}}$ with $F_{\text{MM}}(\psi) = 0$ on it — Type-A [=] by the moment-map construction.

S4 (Variational principle).

The ground energy is $E_0 = \inf_{\psi} \langle \psi | H | \psi \rangle$, and the geometric burden $B_{\text{geom}} = \text{Var}(g_{\text{FS}})$ is bounded below by zero — which supplies the required variational structure.

S5 (discharged).

Metric-flow well-posedness was a primitive in v6.0-F and is derived in v6.5-RF8 from the Hessian identity below; the primitive count is four.

S5' (primitive-vs-emergent, structural).

Every physical observable in RCF is emergent from the relational coarse-graining $C_L : B(\mathcal{H}) \rightarrow B(\mathcal{H})/\sim$, never primitive. A substrate’s primitive-observable algebra $\mathcal{A}_{\text{prim}}$ is admissible as a host for the projection picture only if it admits an emergent interpretation

under some C_L ; the distance $|\mathcal{A}_{\text{prim}} - \mathcal{A}_{\text{emergent}}(C_L)|$ is the substrate’s primitive–emergent gap.

The frontier programme gave S5’ its first positive execution *reading*. The AKLT arc had read the axiom as a *ruling-out* instrument (substrates whose primitives are too distant are excluded); the frontier programme read it as a *construction* instrument: the object the axiom demands — an observable that exists only as an emergent of a relational structure — is precisely what the queued composition instruments are chartered to seek. The audit (Section 7) is the register’s guarantee that no such object enters the record before the machine has measured it: the extension arc’s emergent carrier was narrated without instruments and was struck in full.

2.2 The Hessian identity and the robustness principle

Theorem 2.1 (Hessian identity; Type-A [=] (VN-19, established v6.5-RF8)). *For the Fubini–Study metric on $\mathbb{CP}(\mathcal{H})$,*

$$g_{\text{FS}} = \frac{1}{2} \text{Hess}(\Phi), \quad \Phi = 2 \log \langle \psi | \psi \rangle. \quad (1)$$

Proof. In holomorphic coordinates $[z_0 : \dots : z_n]$ the Kähler potential is $K_{\text{FS}} = \log \sum_k |z_k|^2$; the Kähler metric is $g_{i\bar{j}} = \partial_i \partial_{\bar{j}} K_{\text{FS}}$. Setting $\Phi = 2K_{\text{FS}}$ gives $\text{Hess}(\Phi)_{i\bar{j}} = 2 \partial_i \partial_{\bar{j}} K_{\text{FS}} = 2g_{i\bar{j}}$. $\square$

The identity is a tautology of Kähler geometry; the framework’s content is the *choice* of $\Phi = 2 \log \langle \psi | \psi \rangle$ as the canonical potential of quantum state space. Its structural consequences — the well-posedness of the metric flow (the discharged S5), and the hierarchical RP principle — are unchanged. The principle decomposes robustness into two sequential layers: a Type-A instantaneous layer ($d_t F_{\text{MM}} = -L_F F_{\text{MM}} \rightarrow 0$, rate fixed by the symmetry-algebra spectrum) and a Type-B mechanism layer ($d_t B_{\text{geom}} = -\|\nabla B_{\text{geom}}\|_{g_{\text{FS}}}^2 \rightarrow 0$, rate fixed by the Hessian spectrum), with the detectability contraction $\|\text{error}_k\|^2 \sim \gamma_{\text{DL}}^{2k}$, $\gamma_{\text{DL}} = 1/\sqrt{3}$ (VN-17). The frontier programme does not touch this layer; it supplies, for the first time, a *measured neighbor* of the principle’s deepest commitment — structure that exists only because a relational act occurs: the certified margin field of Section 6 exists only when both channels act, with no single-channel counterpart.

2.3 The epistemic classification

Every vanishing claim in the register carries a tag [1]: **Type-A** [=] — an exact identity forced a priori by symmetry, algebra, or representation theory, constraining the *form* of related quantities; or **Type-B** [$\rightarrow$ rate] — an asymptotic limit whose rate is determined a posteriori by mechanism. The classification’s anti-conflation rules (bridge requirement, mandatory tagging, form-not-rate, default-to-B) govern every claim in this paper. The frontier programme added a third grade family specific to measurement claims — certification grades (Section 4.1) — without altering the Type-A/Type-B spine. Two frontier results are usefully read as endpoint instances of the classification: the Jordan freeze at the relational corner (machine zero, Section 6.2) is an absence measurement backed by the channel’s fixed-point structure; and the zero-ordering law entered as Type-B (measured fits, declared families) and carries its falsifier in the pending N=14 rung and window extensions — the register’s standard route from measurement to law.

2.4 The six-round AKLT arc and the substrate change

The AKLT arc (rounds delta, epsilon, zeta, eta, eta-extreme, Option A; [1, 4]) attempted a concrete realization of the projection picture on the AKLT spin-1 chain under two burden dials ($V_{\text{strain}} = +h \sum_j (S_j^z)^2$ driving the Haldane–large-D transition; $V_{\text{FM}} = -J \sum_j \mathbf{S}_j \cdot \mathbf{S}_{j+1}$ driving the Haldane–Ising transition). The arc’s verdict was negative on every dynamical form the substrate could host, and the verdict is permanent scientific capital: a falsified collapse ordering; a falsified

role-transfer; a characterized conformal floor artifact (the critical L_0 “floor” is the $v\pi/N$ finite-size regularization of the Haldane–large-D critical point, ratio 0.838 against the $1/N$ prediction at distance 0.005); a characterized multiplet artifact (Lanczos within-multiplet splitting; the corrected singlet-channel gap 0.001231 at $J = 0.3$); a complete map of the substrate’s trade-off surface (gap versus information, monotone, no interior extremum, all channels cloaking together at $h \rightarrow \infty$); and the SPT-edge fragility result of Option A (edge splitting follows the bulk gap’s scaling class in lockstep; the edge has no scaling class of its own).

The structural source of the negative verdict was the primitive–emergent mismatch: the substrate hands over spin operators and their correlations as primitives, while the framework requires observables emergent from a relational coarse-graining. Version 7.0 therefore launched the substrate-change fork with a three-condition criterion (primitive–emergent admissibility; nontrivial boundary–bulk decomposition; kinematic rather than SPT-dynamical). The frontier programme is the fork’s first branch: it kept the spin-1 substrate but changed the tier under interrogation — from the substrate’s phases to the *interface between its two books and its two dynamical channels* — and in doing so found, on the same substrate that falsified the projection picture, an interface with measured structure of its own — a margin field, a mesoscopic corner, an order memory — whose *generative* capacity (observables that exist only because of the crossing) is the certified open question. The lesson is registered as the fork’s amendment: the primitive–emergent gap is a property of the *question asked of the substrate*, not of the substrate alone.

3 The frontier substrate and the two-channel model

This section defines the model. Every quantity measured in the programme is defined here, at operator level, together with its dial dependence and its certified window. The definitions are the executable ones: the artifact chain (Appendix C) regenerates every number in this paper from them.

3.1 The chain, the two books, and the kernel

Definition 3.1 (Substrate). The substrate is a spin-1 chain of N sites, $\mathcal{H}_N = (\mathbb{C}^3)^{\otimes N}$, $\dim \mathcal{H}_N = 3^N$, with open boundary conditions. The certified window is $N \in \{6, 8, 10, 12\}$ (statics ladder; the $N=14$ rung is pre-registered and pending, Section 7 and Table 1), $N = 6$ for the joint dynamical cells (the joint-instrument scope declared at R44), and $N \in \{6, 8, 10, 12\}$ for the Lie-channel transport tracks.

Definition 3.2 (The two books). Two commuting-in-book observables partition the substrate’s content:

$$\hat{N} = \sum_{i=0}^{N-1} (S_i^z)^2 \quad (\text{counting book: } m \neq 0 \text{ sites}), \quad (2)$$

$$\mathfrak{M} = \sum_{b=0}^{N-2} P_b^{(2)} \quad (\text{relational book: } S = 2 \text{ bonds}), \quad (3)$$

where $P_b^{(2)}$ projects bond $b = (i, i+1)$ onto the total-spin $S = 2$ sector of $1 \otimes 1 = 0 \oplus 1 \oplus 2$. Both books conserve total S^z ; their spectra are non-negative integers; and their commutator has the certified scale

$$\max|[\mathfrak{M}, \hat{N}]| = \frac{2}{3} \quad (\text{the fifth } \frac{2}{3}; \alpha\text{-invariant, all } N) \quad (4)$$

— the exchange’s enabling condition, machine-exact at $N = 6, 8, 10, 12$.

Definition 3.3 (The kernel and the two vacua). The *relational vacuum* is the kernel of $\mathfrak{M}$: $\varphi_0 \in \ker \mathfrak{M}$, $\dim \ker \mathfrak{M} = 4 = 2^2$ (the valence-bond construction: two virtual edge doublets, deterministic, edge-mode-explicit). The *counting vacuum* is the product state $|0\rangle_C = (|m=0\rangle)^{\otimes N}$. Their book contents are certified exactly:

$$\begin{aligned} \langle \hat{N} \rangle_{\varphi_0} &= \frac{2N}{3} \quad (\text{site RDM} = \mathbb{I}/3 \text{ exactly}), & \langle \mathfrak{M} \rangle_{\varphi_0} &= 0, \\ \langle \hat{N} \rangle_{|0\rangle_C} &= 0, & \langle \mathfrak{M} \rangle_{|0\rangle_C} &= \frac{2(N-1)}{3}. \end{aligned} \quad (5)$$

The kernel's counting profile $\langle \hat{N} \rangle$ over $\ker \mathfrak{M}$ spans the *band* $[N_{k,\min}, N_{k,\max}]$ around $2N/3$ with spread closing exponentially in N (the band-closing law, Section 5.3); the canonical φ_0 sits exactly at $2N/3$ at every N . The mid-chain entropy of the kernel states is $S_{\text{mid}} = \ln 2$ exactly (the information floor), with edge-singlet composites at $\ln 4$.

The constant $c = 2/3$ is the substrate's kinematic signature, and the register holds it as a *family*, not a coincidence: the per-bond kernel content $s = 2/3$; $\langle \hat{N} \rangle_{\varphi_0} = 2N/3$; the commutator scale, Eq. (4); the dynamic landing $B_\infty = c$ (the Jordan channel fills the counting book to exactly the kernel's equilibrium content, R40, machine-exact to 0.05%); and the kernel-profile deviations closing exponentially. One kinematic family, one dynamic landing — the c -suspicion was closed at R41 and has not reopened since.

3.2 The two channels

Definition 3.4 (Lie channel (transport)). The *Lie channel* is the unitary spin-1 XY dynamics

$$H = \sum_{j=0}^{N-2} \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y \right), \quad J = 1, \quad (6)$$

which conserves $\sum_j S_j^z$. It is the substrate's *transport* class: ballistic fronts (free-sector velocity $v = 1.6444 \pm 0.0276$ against $v_{\max} = 2 \min(J, g) = 1.5868$ at the stationary point $k^* = \arccos(g/J)$), and the sole carrier of counting-book interconversion at early times.

Definition 3.5 (Jordan channel (reconciliation)). The *Jordan channel* is the bond-local dissipative damping of the $S = 2$ bond population, with Clebsch–Gordan-declared jump pair on each bond b :

$$L_1 : |2, M\rangle \rightarrow |1, M\rangle \quad (M = -1, 0, 1), \quad L_2 : |2, \pm 2\rangle \rightarrow |1, \pm 1\rangle, \quad (7)$$

at rate dial γ , satisfying $\sum L^\dagger L = \gamma P^{(2)}$ *exactly* (the $S_0 + S_1 + S_2 = \mathbb{I}_9$ completeness of the bond algebra after the A42-1 repair, residual 2.8×10^{-16}). It is the substrate's *reconciliation* class: it flows the counting vacuum into the kernel while filling the counting book to exactly the kernel's equilibrium content ($B_\infty = 2/3$; multi-stage decay $-4.22 \rightarrow \lambda_{\text{eff}} 0.065 \rightarrow \lambda_{\text{tail}} 0.061$; 99.24% decay at $t = 60$; $K_\infty = 0.971$ rising; $|M_z| \leq 8.8 \times 10^{-17}$ throughout).

The two channels are complementary by construction (R39, machine-confirmed in both sectors): transport is Lie-indexed, reconciliation is Jordan-indexed, and the channel split is the substrate's first certified structure. Their composition classes are the two orderings of the joint evolution, below.

3.3 The crossing: joint dynamics and its two orderings

Definition 3.6 (The crossing). The *crossing* is the joint Liouvillian

$$\mathcal{L} = -i[H, \cdot] + \sum_b \sum_{\mu=1,2} \left(L_{b,\mu} \cdot L_{b,\mu}^\dagger - \frac{1}{2} \{ L_{b,\mu}^\dagger L_{b,\mu}, \cdot \} \right), \quad (8)$$

acting on density matrices $\rho(t) = e^{\mathcal{L}t}\rho(0)$ (Lindblad form [5]; integrators cross-validated expm versus 40th-order Taylor to digit agreement, traceA cross-validation 2.8×10^{-15}). The *two orderings* of the crossing are the strobed cycle maps at fixed total exposure, τ per cycle, split s :

$$C_{LJ} = e^{(1-s)\tau\mathcal{L}_J} e^{s\tau\mathcal{L}_H}, \quad C_{JL} = e^{s\tau\mathcal{L}_H} e^{(1-s)\tau\mathcal{L}_J}, \quad (9)$$

with $s = 0.5$ (equal exposures) and $\mathcal{L}_H = -i[H, \cdot]$, $\mathcal{L}_J$ the dissipator of Eq. (7). Their periodic steady states are the PERM instrument's cells (Section 6.4); the order separation between them is the substrate's *order memory*.

Remark 3.7 (The factorised tier). The *upstream* (un-crossed) tier is the product dynamics: the Lie channel alone (e^{-iHt}), the Jordan channel alone ($e^{\mathcal{L}_J t}$), and their statistically independent composition. Upstream the channels are *factorised by definition* — there is no joint measure, only the product of the two channel measures. This is the structural fact that makes any future composition question well-posed: whatever the crossing constructs, the upstream tier has no joint object for it to be a function of — the programme's composition instruments (queued, not yet executed) are chartered against exactly this boundary.

3.4 Dials, windows, and the two frontiers

α (composite dial).

The composite operator $\mathfrak{M}(\alpha) = \mathfrak{M} + \alpha\hat{N}$; ground states $E_0(\alpha; N)$ define the static anatomy. Certified grid $\alpha \in [0.05, 0.30]$ at every rung.

γ (channel-rate dial).

The Jordan rate; certified window $\gamma \in [0.1, 500]$ (scan anchor $\gamma = \mu_1 = 0.413240$, the free-sector gap $2(J - g)$). The two-zeros tail runs $\gamma \in \{30, 100, 300, 500\}$.

N (size dial).

6, 8, 10, 12 certified; 14 is the pre-registered decisive rung of the kill #2 ladder (Section 6.3), pending execution.

τ (clock).

Evolution time; the memory instrument runs $\tau \in \{0.125, 0.25, 0.5\}$ per cycle with saturation windows declared per cell class.

Γ (substrate coupling).

The chart dial of the queued exclusion-chart instrument, with μ_1 the principal book value; linearity of the un-crossed boundary in (Γ, μ_1^2) is the pre-registered expectation, not a certified result (the instrument has not been built).

Definition 3.8 (The two frontiers and the margin). The α -*frontier* (static) is the composite envelope $\text{ENV} - \text{int}(N)$: the PCHIP interpolant of the certified anatomy rows in (N, M) book space, with the kernel face $M = 0$ on $[N_{k,\min}, N_{k,\max}]$ — a strictly convex curve meeting the kernel band at the relational corner. The γ -*frontier* (dynamic) is the NESS occupancy curve: the book-space locus of the non-equilibrium steady states $\bar{\rho}^*(\gamma)$ of the crossed dynamics, certified by its seven-point scan (all *burdened*-NESS; escape margins $\Delta_{\text{env}} = 1.1603, 1.1582, 1.1523, 1.0962, 0.9005, 0.5835, 0.1935$ at $\gamma = 0.1, 0.3, 0.41324, 1, 3, 10, 30$ respectively — strictly monotone, no crossing). The two are *distinct curves* meeting only asymptotically at the relational corner — R42's certified two-frontier structure. The *margin field* is the escape of the joint dynamics above the static frontier:

$$\Delta(t) = M(t) - M_{\text{env}}(N(t)), \quad (10)$$

measured on trajectories from either vacuum. The margin is the programme's central observable: everything after R42 is either its zeros, its saturation structure, its order-sensitivity, or (queued, unexecuted) its decomposition.

The NESS scan itself carries the R39–R40 capacity structure: the NESS at declared rates fails on both sectors (free-sector asymmetry $18.69\times$; interacting, replicated), all operating points are burdened, $\gamma^* > 30$ (balance extrapolation ~ 35), and n^* climbs $0.595 \rightarrow 0.662$ toward $c = 2/3$ as γ grows — the equilibrium burden profile is the Jordan channel’s attractor, dynamically selected rather than chosen.

4 The verification apparatus

The frontier programme’s results are only as strong as the instrument discipline that produced them, and that discipline is itself part of the framework — the register treats the apparatus as infrastructure with its own certified invariants. This section specifies it, and specifies with equal care what happens when the discipline is *absent*: the audit of Section 7 is the apparatus’s own negative control.

4.1 Certification grades

Every number in this paper carries a declared grade: *machine-identity* ($\leq 10^{-12}$ against a certified counterpart; the bit-exact class is stronger still — byte-level reproduction across fresh re-executions); *statistical* (a fit with declared rmse, window, and family); *absence-grade* (a verified null at a declared resolution floor); and *citation* (a value certified in a prior round and imported by hash, not re-run). Windows are declared per cell class (headroom fields on every sheet since F-V4-5); tolerances are chosen, not inherited; and every quoted value regenerates under a *named extractor* (the quote-with-convention rule, V-4). A fifth status was added by the audit: *quarantined* — a claim that exists in the record without an artifact; such claims are never cited as support (Section 7).

4.2 Pre-registered sheets and pre-commitment

Round sheets are pre-registered before execution (the print order is the seal), keyed by a sheet hash under which all checkpoint cells are stored; imports cite the generating sheets’ hashes and the generator scripts’ md5, byte-compared at import. The certified chain’s sheets are: the R41 composite α -scan (sheet hash 4a75ae26d1, generator md5 49b0ad8cfbe5), the R42 frontier cells (6bb4f559cf), the R43 census (b3dbc822ed), and the R44 N-ladder (c44f3d1192). The binding conditions promoted from interlude findings (F-V4-1 full-precision bisection keys, F-V4-2 pre-registered extractors, F-V4-5 headroom fields) are contract clauses of the R44 sheet, and the R44 sheet’s pre-registered clauses — the KAPPA-c extractor, the kill #2 firing clause, the L-44 lean clauses, the two-zeros families with resolution floor $\varepsilon_0 = 10^{-8}$ — are the programme’s genuine quantitative pre-commitments (Appendix B).

One apparatus claim carried by the superseded v8.0 does not survive the audit: a ten-entry sealed-prediction ledger (A–J) with eight confirmations, one kill, and two pending seals. No seal text, hash, or timestamp for A–J exists in any sheet; the ledger is struck in full (Appendix B, Table 3). The register’s honest note is the mirror of the original one: the certified chain confirmed its pre-registered clauses where the instruments could already constrain answers, and the discipline’s first genuine *kill-by-artifact* remains ahead of the record, not behind it.

4.3 The frontier property census and the proposed crossover taxonomy

The certified census object is the R43 *frontier property census* (V-6-verified): the margin field’s containment structure (eleven positive escape margins, strictly monotone, no crossing), the approach law (tail slope -1.4310 , scope corrected to crossover), the corner geometry ($N_c = 4.0005$, entry at $\gamma = 100$, ray angle steep and nearly fixed), the J-dial response family, and the BAL-q drive table. It is the instrument inventory from which every later claim about the frontier’s properties must draw.

A *crossover-difference census* — a classification of every registered observable by what crossing the frontier does to it — was proposed during the (unexecuted) extension arc, with classes F0–F6: identity preserved; value shift; behaviour shift; identity split; emergent; combination; annihilation. The taxonomy is retained here as a *proposed* instrument charter only: its class table is well-formed, its falsifiers are statable, and none of its rows have ever been executed against data. Any future census execution must pre-register its rows and extractors in a sheet before the first cell runs, under the rules that governed the R43 census itself.

4.4 The defect taxonomy

The verification interludes name their own defects, in the round, before adjudication. Across the executed chain the defects fell into three families, with zero new families — and the audit added a fourth:

1. **Instrument defects** (F-V4-1: bisection keys whose 6-decimal round-trip does not pin the mid at which a value was computed, a latent same-key collision hazard quantified at 2.5%; F-V4-2: quoted anchors whose extractor convention was not pre-registered — the dead κ anchors 0.14/0.117/0.106; F-V4-4: quote-precision drift in narrative numbers). Structural fixes: full-precision keys, extractor fields on every anchor, quote-with-convention — all promoted to binding sheet conditions and verified working on the R42–R44 sheets (the zero-findings result of V-5).
2. **Auditor-side defects** (V6-D1: incomplete declared-exception list in the verifier’s own source-scan; V6-D2: a phrase check inheriting the wrong round’s wording; V6-D3: a budget window exceeded mid-fresh-block). All three were caught by the verifier’s own discipline, repaired in place, and re-run; zero defects were found in the audited sheets’ data paths.
3. **Environment defects** (F-V4-5: memory-marginal cells whose certified runs predate a re-boot; the remedy — headroom fields, bit-exact prefix verification, and second-method cross-validation at 2.2×10^{-15} — is now a binding condition).
4. **Verification without artifacts** (A-1, this round): claims and verdicts — including entire verification interludes — narrated in the record with no script, checkpoint, or log behind them. This family is the gravest of the four: it corrupts the register’s trust structure rather than a single value. The structural fix is the audit itself (artifact-gated citation), and the family’s standing rule is stated in Section 7.5.

The taxonomy’s stability through the executed chain, and the audit’s clean separation of that chain from the artifact-free arc, is load-bearing for this paper: an apparatus whose failures are predictable in class, and whose record can be shown to contain nothing that the machine cannot regenerate, is one whose confirmations mean something.

5 The static frontier

5.1 The composite anatomy and the deficit law

The static instrument diagonalizes the composite $\mathfrak{M}(\alpha) = \mathfrak{M} + \alpha\hat{N}$ (the $S^z = 0$ block is exact by symmetry: both books conserve S^z and the composite ground states live in $S^z = 0$; the block spectrum equals the full spectrum to 1.1×10^{-14} at shared rows). The composite ground envelope shows *no crossover*: every $\alpha > 0$ grid row is intermediate (the kernel holds the ground only at $\alpha = 0$ exactly), and both pre-registered critical- α candidates are dead. The parabolic corner-cutting is quantified by the deficit law — the excess of the counting line over the composite ground:

$$D(\alpha; N) = \alpha N_{k,\min}(N) - E_0(\alpha; N) = C(N) \alpha^{\beta(N)}, \quad (11)$$

fitted on the pre-registered grid $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ at every rung (extractor KAPPA-c, declared before first execution under the F-V4-2 binding condition; the dead anchors 0.14, 0.117, 0.106 of the pre-ladder era are not reused; the N=12 rung extends its grid to eleven alphas, $\alpha \leq 2.0$, all stored in the checkpoint). The certified ladder: $\beta = 1.9758, 1.9736, 1.9727, 1.9509$ at $N = 6, 8, 10, 12$. The law is *not* an exact two-exponent: the weak confirmation L-44 (clause iii failed at $\sim 2\sigma$ on the last rung) registers the deficit power's own N-structure — the same finite-size softness class as the curvature law's drift, below.

$D(\alpha; N) = C(N) \alpha^{\beta(N)}$ with β drifting toward 1.95 across $N = 6$ –12; fitted per rung at log-space rmse ≤ 0.0098 ; falsified as an exact α^2 at the last rung. The drift is itself registered: the parabolic corner-cutting is N-soft.

5.2 The curvature law and the mesoscopic corner

The curvature of the composite frontier at the relational corner — the Legendre dual of the deficit law — is $\kappa(N) = 1/(2C(N))$: near the corner, $M_{\text{env}}(N') \sim (N_{k,\min} - N')^2/(4C)$. The certified ladder:

N	6	8	10	12
$\kappa(N)$	0.2938	0.2497	0.2138	0.1932
$\beta(N)$	1.9758	1.9736	1.9727	1.9509

$\kappa(N) \propto N^{-\gamma_C}$ with $\gamma_C = 0.6131 \pm 0.0187$ (four-point fit; re-derived by the audit from the stored κ ladder to 0.6131, SE 0.0187). The corner's *class* is mesoscopic: γ_C lies two standard errors from both the edge class (0) and the bulk class (1). Corner width $\sim N^{0.31}$ in total book units — a finite, well-defined mesoscale object. The law's own drift is registered: the corner is N-soft, not N-hard.

The mesoscopic class verdict is the frontier's first spatial-class certificate, and it is the anchor the pending questions lean on: a genuinely intermediate object, with its own width exponent and its own decay exponent. The audit's pre-registered prediction for the pending N=14 rung (Table 1) is that the rung lands on the extrapolation $\kappa(14) = 0.1754$ — a test of the law, not a member of it.

5.3 The band-closing law and the non-echo certificate

The kernel band closes exponentially — the counting spread of $\ker \mathfrak{M}$ around $2N/3$ decays as $\varepsilon(N) = 2N/3 - N_{k,\min}(N) \sim e^{-N/\xi}$ with $\xi = 1.0467$ sites (closing ratios 0.148/0.139/0.133, mildly accelerating; the audit's extrapolation for the pending rung is $\varepsilon(14) \approx 4.4 \times 10^{-6}$). The static projection test then separates the two objects decisively:

$$\kappa/\varepsilon = 26.7 \rightarrow 153.6 \rightarrow 947.0 \rightarrow 6417.2 \quad (N = 6 \rightarrow 12; \times 240 \text{ across the ladder}). \quad (12)$$

The curvature is *not* the band edge's echo: the band closes exponentially, the curvature decays as a power — two objects, two N-laws. The static-projection fork (frontier as the books' projection at the corner versus frontier as independent object) adjudicates on the independent-object side, and the margin's γ -side behaviour (law 3) is the same verdict reached from the dynamic tier: what the projection lacks, the two-zeros law proves exists.

5.4 The three-way zero ordering

The decomposition-completeness question — does transport account for all of the margin? — was sealed as a fork with pre-committed families and a declared resolution floor $\varepsilon_0 = 10^{-8}$. On the γ tail $\{30, 100, 300, 500\}$:

γ	30	100	300	500
margin Δ	0.1935	0.0243	0.00287	0.00104
drive T	0.2741	0.1047	0.0364	0.0220
$\Delta/ T $	0.706	0.232	0.0788	0.0475

POWER fits win decisively over stretched-exponential and contact alternatives (log-space rmse 0.0286/0.0208 versus 0.31/0.13 and 0.27/0.12): $\Delta \sim 114.8 \gamma^{-1.859}$, $T \sim 6.10 \gamma^{-0.899}$, ratio exponent -0.9606 ± 0.0166 (-0.96 at 2σ from zero; re-derived by the audit). Floor crossings at $\varepsilon_0 = 10^{-8}$: $\gamma_\Delta = 2.6 \times 10^5$, $\gamma_T = 5.9 \times 10^9$, with the projected zero bracketed $\gamma_{\text{proj}} \in (30, 100)$:

$$\gamma_{\text{proj}} < \gamma_\Delta < \gamma_T, \quad (13)$$

robustly separated across three orders of magnitude. The margin dies roughly twice as fast as its transport driver; the decomposition is **incomplete** — **non-transport carriers exist**.

The ordering’s composition-level attribution — which carrier owns the margin’s faster death — is exactly the queued decomposition question; the audit quarantined one narrated answer to it (Section 7) and the instrument remains to be built.

5.5 Transport saturation: the N-side half of the same verdict

The transport-only margin family (Lie channel alone, from φ_0) saturates per bond:

$$\Delta_T(t=5)/(N-1) = 0.0529 \rightarrow 0.1500 \rightarrow 0.1489 \rightarrow 0.1468$$

across $N = 6 \rightarrow 12$ — the $N = 6$ cell is finite-size, and the per-bond transport margin is a *bulk object* with a flat, N -independent profile. Consequence: the *N-side transport zero does not exist*. Together with the γ -side ordering, the zero structure closes symmetrically: on the γ axis the margin dies at twice the transport rate (non-transport content exists); on the N axis transport saturates while the margin’s own ladder rises toward its $N = 6$ level (Section 6.3) — the extrapolated zeros of the pre-census era are projection artifacts, and the margin is a non-transport object whose measured composition is queued.

6 The dynamic frontier

6.1 The two-frontier separation

R42 certified the dynamic frontier as a tier of its own. All seven NESS points sit strictly off the static frontier (Δ_{env} strictly monotone in γ , no crossing, clause margins 4–23 $\times$ the floor), and the lean L-42 (“frontier-maker”, declared with bias flag on) was **confirmed-strong** at 7/7. The two frontiers — the α -frontier of the algebra tier and the γ -frontier of the joint tier — are distinct curves in book space, meeting only asymptotically at the relational corner. The escape-margin field between them is the first measured content of “the substrate contributes”: the margin of Eq. (10).

6.2 Channel typing

The sharpened question of R42 — corner-mover or frontier-maker? — resolved as a *channel-typed split*, and the split itself is the datum:

Channel	Typing	Certified reading
Jordan (from φ_0)	corner-keeper	frozen on the kernel face at machine zero ($d_{\text{env}} = -4.2 \times 10^{-17}$; all X-rates void)
Jordan (from $ 0\rangle_C$)	corner-selector	attractor ON the kernel face (landing +0.0253 at $T = 60$)
Joint	frontier-maker	attractor +1.10 off the static frontier at the operating point; kernel-band crossing at $t = 1$ (the minting overshoot datum)
Lie (from φ_0)	slope-breaker	realized -1.03 in transit versus the slope field -0.24
Lie (from $ 0\rangle_C$)	on-slope trader	initial trade rides the frontier's local slope (+0.003 mismatch at $t = 0.5$), then departs

The freeze at the relational corner is certified static as well: the G3 void at 10^{-16} — the relational vacuum is the Jordan channel's fixed point. Front-level structure is similarly typed: the two books' edge-mid gaps disagree at $t = 0.5$ and $t = 2$, agree at $t = 0.25, 1, 5, 20$ (the observable-class SPLIT cell), and the T' -arrival table types the orderings: Lie flat; joint edge-lagged by 0.300; Jordan edge-lagged by 0.500 — amplitude overshoot without front lead.

6.3 Kill #2: the margin ladder and the pending rung

The margin's N-side ladder is the kill #2 instrument. The transport-accounting ratio $X(N) = |X-t_1(N)|$ reads 1.0306, 1.0179, 1.0223, 1.0294 at $N = 6, 8, 10, 12$: the dip is at $N = 8$, the second half rises monotonically toward the $N = 6$ level ($0.0223 \rightarrow 0.0294$), and the rung survives its window by 0.0012 (4%; $d_{12} < d_6$).

The pre-registered firing clause was mechanical: not-fired iff net decay and second-half envelope strictly inside first-half. The clause read not-fired at $N = 12$ by 0.0012 with a rising tail — the macro-limit claim survives its second window, and the shape is registered honestly. The decisive rung is $N = 14$ (the statics side is cheap: BLOCK-e, the $S^z = 0$ -restricted composite anatomy, is exact by symmetry; the transport rung needs the restricted Lie track). No $N=14$ cell has executed; no adjudication rule beyond the standing clause has been sealed; the extrapolation landscape is published in Table 1 so that the rung's reading is a test, not a choice.

The ladder's honest tension is structural and worth carrying forward explicitly. On the γ axis the margin dies at twice transport's rate (law 3); on the N axis the same object's ladder rises back toward its $N = 6$ level while per-bond transport saturates flat. The two axis-profiles of one object are sign-opposed, and the resolution — mixture, turn, or persistence — is exactly what the pending rung and the queued decomposition instrument are for. What the certified record rules out is the projection-only reading on both axes: a projection artifact has no mechanism for dying faster than its driver on one axis while re-bounding on the other.

6.4 The order memory and the margin's order-sensitivity

R44's PERM instrument measured the crossing's response to channel order, and the response is machine-real at three independent levels:

1. **The order-memory slopes.** The $\tau \rightarrow 0$ limit of the book separation between the two orderings, $\kappa_{\text{perm}} = \lim_{\tau \rightarrow 0} [\text{books}(C_{LJ}) - \text{books}(C_{JL})]/\tau$, is nonzero at machine significance

for the full slope vector $(\kappa^M, \kappa^N, \kappa^K, \kappa^{\text{margin}})$:

$$\gamma = 1 : (-0.0155, +0.0951, +0.0074, -0.0097),$$

$$\gamma = 3 : (-0.0336, +0.2135, +0.0192, -0.0221),$$

— the order-memory vector points in the grip’s canonical book quadrant $(-M, +N)$ and grows $\times 2.22$ from $\gamma = 1$ to 3.

2. **The steady pair.** The periodic steady states at the operating point ($\gamma = 1, \tau = 0.5$) separate:

$$(M, N, K, \Delta_{\text{env}}) = \begin{cases} (1.0798, 3.6925, 0.1481, 1.0667) & \text{Lie-first,} \\ (1.1531, 3.6501, 0.1335, 1.1369) & \text{Jordan-first,} \end{cases}$$

— a margin difference of 0.070, which is 6.5% of the margin.

3. **The protocol-invariance split.** Against the census NESS curve (class 5×10^{-3}): the Lie-first landing sits on-curve (residual $+0.0032$); the Jordan-first landing sits off-curve ($+0.0129$). The frontier curve is protocol-invariant under the Lie-first ordering and is *not* reached by the Jordan-first ordering: the margin is order-sensitive — the frontier-resident object responds to channel order.

The kernel overlap responds to order as well ($dK > 0$ under the one-block family at every measured (γ, τ) — the back-coupling candidate’s first machine reading), and the full state-space battery that would adjudicate it is queued (Section 9). Together these three levels constitute the substrate’s first certified *memory*: the crossed dynamics does not merely forget its assembly order in the steady state; it retains a signed, book-structured, drive-growing residue of it.

7 The verification audit (A-1) and the quarantined extension arc

This section is new in v8.1, and it is the reason v8.1 exists. The director’s mandate was explicit: re-read the programme’s full history, machine-verify every calculation and value it carries, and correct the record so that no false prediction, calculation, or speculation is carried forward. The audit was executed as a script (`scripts/audit_fw81.py`; log `RCF_v81_audit_log.txt` under `download/`), and its findings are the section’s content.

7.1 The artifact census

The audit’s first act was a census, not a computation: for every round and interlude the programme’s record claims, does an artifact exist — a script, a hash-keyed checkpoint, a run log? The result partitions the record cleanly:

Segment	Artifacts	Status
R39–R44, V-1–V-6	scripts, checkpoints (md5-verified against certified backups), run logs, verification harnesses	machine-held
R44’s interlude V-7	none (the verifier was never written)	standing debt, partially discharged here at the value level
R45–R50, V-7–V-13	<i>zero scripts, zero checkpoints, zero logs, anywhere in the project</i>	conversation-only — struck

The five checkpoints of the executed chain are byte-identical to their certified backups (md5 `d060254a...`, `b816a39a...`, `9730652c...`, `76a45f04...` for R40–R43; R44’s checkpoint `b7340513...` is intact and is the object of this section’s re-derivations). The extension arc’s absence is total: no generator, no sheet, no result cell, no verification manifest — nothing a machine could re-run.

7.2 The certified chain: value-level re-verification

The audit then re-verified every quantitative claim the certified chain carries into this paper — fifty-eight checks, all PASS against the checkpoints. Beyond value-matching, the audit re-derived the R44 fits from the raw rows stored in the checkpoint (four years of narration could not have drifted these numbers, and did not):

- the four KAPPA-c deficit fits (β , C , κ , rmse at every rung, on each rung’s full stored grid) reproduce to 0.02%;
- the curvature law re-fits to $\gamma_C = 0.6131$ (SE 0.0187) from the stored κ ladder;
- the two-zeros POWER fits re-derive ($\Delta \sim 114.8 \gamma^{-1.8594}$, $T \sim 6.10 \gamma^{-0.8989}$, ratio exponent -0.9606 , floor crossings 2.6×10^5 and 5.9×10^9);
- the kill #2 ladder, the PERM steady pair, and all eight κ_{perm} slopes re-derive (the slopes from the stored one-block deltas under the sheet’s own origin-constrained LSQ convention, machine-exact);
- the R42 channel-typing cells, the R43 census values ($N_c = 4.0005$, $p_N = -1.4310$, $p_M = -1.5804$), the NESS tables, and the R39/R40 anchor values (front velocity, capacity asymmetry $18.69\times$, B_∞ , decay rates) all match their logs.

One debt is registered rather than hidden: R44 never received its own verification interlude (V-7 was narrated but not written). The audit’s re-derivations discharge that debt at the value level — the checkpoint’s stored values are internally consistent and regenerate under independent refits — but the bit-exact fresh-execution class of the V-1–V-6 pattern remains owed to R44 by the next interlude that runs.

7.3 The quarantined arc: claim inventory

The extension arc (R45–R50, V-7–V-13) is struck from the certified record. Its full claim inventory is retained — every quantitative claim, with the audit’s verdict — in Table 3 (Appendix B), because a retraction that does not name what it retracts protects nothing. The verdicts are of four kinds:

Arithmetically consistent, unmeasured.

The arc’s carrier decomposition $\Delta = C + L + J + \chi = 0.0302$ with $C/\Delta = 0.551$, $\chi/\Delta = 20.5\%$; the lever identity $0.551 \times 4.9\% \times 0.0302 = +0.0008$; the summed exponent $(-1.118) + (-0.743) = -1.861$. All internally exact, all without a single measured input. Consistency is not evidence: a self-consistent number system with no instrument behind it is a story, and the register does not carry stories.

Extrapolation-consistent.

The arc’s $\kappa(14) = 0.1754$ equals the certified four-point log-log extrapolation (0.17535) to 0.03% — the claim is what the certified law *predicts*, which is exactly why it is re-registered here as a prediction (Table 1) rather than discarded.

Inconsistent with certified data.

Four demonstrations, below.

Unverifiable apparatus.

The interludes V-7–V-13 (with claimed cell counts 31/31 through 43/43), the sealed-prediction ledger A–J (eight “confirmed”, one “killed”, two “pending”), the self-application ledger “15:7:1” (the certified ledger stands at 8:7 through R44), and the twenty-three-row crossover-difference census: none of these exist as artifacts. The claimed verification *of the*

arc's own claims is the arc's gravest member, because it is a verification claim — the one class of claim whose truth the register can check without running physics, and it failed the check.

7.4 Four demonstrated inconsistencies

The arc's $X(14) = 1.0302$ — the reading on which “kill #2 closed by sealed rule” and the entire growth-lever edifice rested — matches no extrapolation of the certified ladder (linear 1.0252; cubic 1.0249; quadratic 1.0499). Its step arithmetic also conflates two conventions: the -0.0012 is $d_{12} - d_6$ (four rungs), the $+0.0008$ is $d_{14} - d_{12}$ (two rungs). Kill #2's certified status is unchanged: *not fired at $N=12$ by 0.0012; the $N=14$ rung pending*.

The arc's $\varepsilon(14) = 1.55 \times 10^{-6}$ and “local scale 1.026” contradict the certified band-closing law ($\xi = 1.0467$, closing ratios 0.148/0.139/0.133 accelerating): the law extrapolates to $\varepsilon(14) \approx 4.4 \times 10^{-6}$ (next closing ratio ≈ 0.128 , local scale ≈ 0.97). The quoted values imply a closing ratio of 0.0515 — a $3.9\times$ acceleration with no certified mechanism and no artifact.

The arc attached itself to a certified number by equivocation. R44 certified a steady-state order sensitivity of 0.070 — 6.5% *of the margin* (1.067). The arc's identity “6.5% = $2\chi \times f_\tau$ with $f_\tau = 0.158$ ” evaluates to 0.00196 — 0.20% of the margin, a factor ~ 36 off the certified value. The identity holds only if “6.5%” is silently re-normalised to 6.5% *of $\Delta = 0.0302$* — the arc's own unmeasured quantity. The companion claim (“the JL off-curve residual $+0.0129$ is 2χ ”) is two-digit post-hoc matching of an unmeasured number to a certified residual.

The arc's “two-component refit” ($\chi \sim \gamma^{-1.861}$, substrate $\sim \gamma^{-0.518}$; “rmse 0.029 $\rightarrow$ 0.007”) fits *four* free parameters to the certified tail's *four* points — a saturating fit with zero statistical content. Worse, its curvature is backwards: the certified tail's local exponents steepen ($1.72 \rightarrow 1.94 \rightarrow 1.99$), while a fast-plus- slow two-component sum necessarily flattens. And the arc's -1.861 is silently interchanged with the certified -1.8594 elsewhere in its own narrative, each presented as an exact “0.1% match” to the other.

7.5 Disposition, and the pre-registered N=14 predictions

The disposition rule the audit installs is one sentence: *no number enters the certified register without an artifact, and every struck claim is re-registered as a prediction or as nothing*. Applying it:

- the arc's **apparatus** (V-7–V-13, seals A–J, the 15:7:1 ledger, the executed census) is struck outright — Table 3;
- the arc's **unmeasured physics** (carriers, torus, chart, formation clock, listening law) is struck from this paper's certified body and survives only as the quarantined inventory of Table 3, each row available to future sheets as a *pre-registered hypothesis* should the director choose to run it for real;
- the arc's **testable content at the pending rung** is converted into the prediction table below, anchored to certified extrapolations where they exist.

Table 1: Pre-registered predictions for the pending N=14 rung (the audit’s conversion of the arc’s testable claims). Values are certified-ladder extrapolations, not measurements; the rung’s reading is a test of them.

Quantity	Prediction	Basis and falsifier
$\kappa(14)$	0.1754	four-point log-log extrapolation (0.17535); falsified if the rung deviates beyond the fit’s SE band
$\beta(14)$	1.95 ± 0.01	the N-soft drift $1.976 \rightarrow 1.951$; an exact α^2 reading is already falsified at $N = 12$
$\varepsilon(14)$	$\approx 4.4 \times 10^{-6}$	the exponential closing law ($\xi = 1.0467$); a “band echo” at 1.55×10^{-6} is falsified by this table
$X(14)$	undetermined	no certified extrapolation is stable (linear 1.0252 / cubic 1.0249 / quadratic 1.0499); the kill #2 clause reads the rung mechanically once measured
γ_C refit	0.614 ± 0.016	five-point refit of the curvature law; the mesoscopic class must survive its third standard-error test

The audit closes with the register’s own words for what happened. The programme’s discipline was built so that claims which outrun their instruments die at verification; the arc’s claims outran their instruments *and* their verification, and survived only because the verification itself was narrated. The A-1 audit is the machine’s answer to exactly that failure mode, and its existence is the reason the certified chain’s confirmations still mean something: an apparatus that will retract twenty-five rounds of attractive narrative rather than carry one unmeasured number is an apparatus whose kept numbers are load-bearing.

8 Synthesis: the certified register

8.1 The law register

The certified laws of the frontier programme, with their grades and falsifiers. Every entry is machine-held or audit-re-derived; nothing in this table rests on a narrated round.

#	Law	Certified content	Grade / falsifier
1	deficit law	$D = C(N)\alpha^{\beta(N)}$, $\beta \rightarrow 1.95$ over $N = 6-12$ (N-soft; falsified as exact α^2)	Type-B; log-rmse ≤ 0.0098
2	curvature law	$\kappa \propto N^{-\gamma_C}$, $\gamma_C = 0.6131 \pm 0.0187$; corner width $N^{0.31}$; mesoscopic class	Type-B; five-point refit pending (Table 1)
3	zero ordering	$\gamma_{\text{proj}} < \gamma_{\Delta} = 2.6\text{e}5 < \gamma_T = 5.9\text{e}9$; ratio exponent -0.96	Type-B; POWER rmse 0.029/0.021; non-transport carriers exist
4	transport saturation	per-bond $\Delta_T \rightarrow 0.147$ for $N \geq 8$; the N-side transport zero does not exist	Type-B; flat-profile falsifier: any rung off 0.147 by > 5%

#	Law	Certified content	Grade / falsifier
5	two-frontier separation	seven NESS points strictly off the static frontier, monotone, no crossing; clause margins 4–23× the floor	Type-B + absence; any crossing falsifies
6	channel typing	the relational vacuum is the Jordan channel’s fixed point (machine zero; X-rates void); Lie is the slope-breaker; joint is the frontier-maker	absence + structural; any Jordan X-rate at φ_0 falsifies
7	order memory	$\kappa_{\text{perm}} \neq 0$, vector $(-M, +N)$, growth $\times 2.22$ ($\gamma 1 \rightarrow 3$); margin order-sensitive 6.5%; on-curve only under Lie-first	Type-B; the full state-space battery is the queued adjudication

8.2 The compressed history

Round	Established
R39	first data round: channel split machine-confirmed (transport Lie-indexed, reconciliation Jordan-indexed); NESS failure at declared rates (capacity asymmetry 18.69×)
R40	two-convention vacuum fork: the two vacua are two different zeros; the Jordan channel selects the equilibrium burden profile ($B_\infty = 2/3$); V-3 registered the cache-staleness defect class with verdict-neutral corrections
R41	the α -frontier exists (static convexity certificate); the exchange is a field, not a rate; the c -suspicion closes; the convention-hash regime’s first full cycle
R42	the γ -frontier is a distinct curve (L-42 confirmed-strong); channel typing (keeper / selector / maker / breaker); kill #4 does not fire
R43	the frontier property census: margin field, corner geometry, approach law, dial family (V-6-certified)
R44	composition: curvature law, mesoscopic class, the two zeros separated, the order memory, the K-response’s first reading
A-1	this audit: the executed chain re-verified (58 checks); the unexecuted extension arc struck and quarantined; the N=14 predictions pre-registered

8.3 What the frontier is, on the certified record

The programme’s founding question — is the interface between the substrate’s algebraic tier and its dynamical tier more than a boundary? — has a certified answer of a specific, restrained shape. The interface is a *field-bearing object*: it carries a margin that dies faster than its transport driver on one axis and rebounds on the other; it owns a mesoscopic corner with its own width and decay exponents; it is the unique place where the substrate’s two books and two channels produce behaviour irreducible to any single channel (the order memory, the protocol-invariance split). The un-restrained version of the answer — a generative interface constitutively producing new observables — is exactly the part of the story that did not survive the audit: it is the open question, not the finding.

The honest ledger at programme close: kills #1 and #3 standing not-fired; kill #4 not fired (R42); kill #2 not fired at $N = 12$ with the decisive rung pending; leans L-41 refuted, L-42 and L-43 confirmed-strong, L-44 confirmed-weak; self-application ledger 8:7; chain $R39 \rightarrow R44$, six links, machine-held or debt-cleared — and one arc struck.

9 Open problems

The programme’s residue is small, named, and every item is instrument-ready: each names its extractor, its budget class, and its pre-registered reading. The queue, in the register’s priority order:

The N=14 rung (kill #2’s decisive point).

The statics side is cheap (BLOCK-e, the $S^z = 0$ -restricted composite anatomy, exact by symmetry; the R44 probe validated it to 10^{-14} on shared rows); the transport side needs the restricted Lie track. The pre-registered predictions are Table 1: the rung’s reading closes kill #2 mechanically, tests the curvature law’s five-point refit, and settles the “turn” question — whether the margin’s N-side ladder reverses into growth or continues its envelope-constrained rise.

The R44 verification interlude (V-7, standing debt).

R44’s checkpoint has no certified backup because its verifier was never written. The audit’s re-derivations hold at the value level; the bit-exact fresh-execution class of V-1–V-6 is owed before the chain extends further.

The margin decomposition instrument.

Law 3 certifies that non-transport carriers exist; nothing yet measures them. The queued design: the block/excision cross (corner-mask excision with zero-excision baseline identity; channel blocking with zero-block identity), executed as a sheet with pre-registered additivity nulls — the quarantined arc’s carrier story (Table 3, rows 1–6) is available as the hypothesis this instrument would test, and only as that.

The κ_{perm} state-space battery (the Q4 adjudication).

The order memory is certified at three levels (Section 6.4); its full state space — the order family beyond the one-block class, the steady-level τ -family, the kernel response dK ’s mechanism — is the register’s next physics word. The narrated “response torus” and “listening law” of the struck arc (Table 3, rows 7–14) are re-registered there as pre-registered hypotheses, each with its falsifier statable in advance.

The $\Gamma = \mu_1$ exclusion chart.

The queued instrument for the substrate’s own boundary document; the un-crossed boundary’s linearity in (Γ, μ_1^2) is the pre-registered expectation. The struck arc’s “palimpsest” reading is quarantined as hypothesis, not cited.

The joint instrument beyond $N = 6$.

The margin field’s N-scaling beyond the operating point; the named path is the $q = 0$ -block Liouville reduction. The two-zeros tail’s window-extension caution (extrapolated zeros are projection artifacts until measured) governs every extension.

Cylinder 2D transport and conservation source terms.

The standing pre-frontier queue items, unchanged.

Two standing questions of the pre-frontier register remain open in their original form: the projection picture’s dynamical forms (falsified on this substrate; the kinematic standard-GR cloaking asymmetry survives as scoped in v7.0), and the substrate-change fork’s remaining branches — now informed by the frontier programme’s amendment that the primitive–emergent gap is a property of the question asked of the substrate, not of the substrate alone. To these the audit adds its own entry: the apparatus’s fourth defect family (verification without artifacts) has been closed by retraction this time; the register’s standing obligation is that it never needs closing that way again.

10 Conclusion

The frontier programme began with a question the framework had carried since its founding: whether an interface between an algebraic tier and a dynamical tier could be more than a boundary — and if so, what. Six executed data rounds and six machine verification interludes later, the certified answer is specific and restrained: the interface exists as a measured object — a margin field with sign-opposed axis-profiles, a mesoscopic corner with its own exponents, an order memory signed into the books — and whether it is *generative*, the property the framework’s founding axiom most wants from it, is the open question the queued instruments own.

The programme’s decisive event, in the end, was not a discovery but a retraction. Between R44 and this document, an extension arc of twelve narrated rounds — a complete physics of interference carriers, response tori, and inscribed boundaries — accreted onto the record without a single artifact, including narrated verifications of its own claims. The audit struck it: fifty-eight checks re-held the executed chain, four demonstrations pinned the arc’s contradictions against certified data, and the arc’s testable content was converted into pre-registered predictions for the pending rung. The framework’s discipline was designed on the premise that claims which outrun their instruments must die at verification; this audit is the first time the premise was applied to the register’s own narration, and the register survived it — smaller, harder, and trustworthy in the only sense that matters here: every number that remains can be regenerated by a machine that does not care about the story.

What the framework now holds, beyond v7.0: a certified composite frontier with a mesoscopic corner (laws 1–2); a zero structure that proves the interface carries non-transport content (laws 3–4); a two-frontier geometry with typed channels (laws 5–6); and the substrate’s first certified memory (law 7). What it does not hold — and this is the document’s most load-bearing sentence — is every quantity whose only provenance was conversation: the carrier χ and its decomposition, the summed-exponent law, the formation clock, the torus, the inscription, the seals A–J. Those are quarantined hypotheses with statable falsifiers, awaiting instruments that have not yet been written.

The chain is closed, audited, and honest. The next word is the N=14 rung.

References

References

- [1] Z.ai | Super Z collaboration, *RCF-FW-v7.0: The Relational Coarse-Graining Framework, Post-Arc Rewrite* (2026). Internal framework document; lineage v6.5-Protocol#20 + v6.5-RF8 + the six-round AKLT arc. Artifact: `upload/RCF-FW-v7.0.pdf`. (Historical phase title; see Ref. [2] for the canonical-name resolution.)
- [2] Z.ai | Super Z collaboration, *AM-15: canonical framework-name resolution* (2026). Ratified Rule-Zero amendment: the canonical title across all active development is *Relational Constraints Framework* (RCF, plural); historical forms — Relational Constraint Framework (Gen-2 manuscripts), Reconciliation Causal Framework (Phase B), Relational Coarse-Graining Framework (v7.0), Relational Complexity Framework (v7.1 subtitle) — are indexed as phase-specific aliases. Artifact: `upload/Pasted_Content_1788081965390.txt`, Deliverable 2.
- [3] Z.ai | Super Z collaboration, *RCF-FW-v8.1 verified-change basis: the frontier programme artifact chain* (2026). Rounds R39–R44, interludes V-1–V-6, and the A-1 audit; scripts `r39_stageb_free_sector.py`, `r40_vacuum_fork.py` (generator md5 49b0ad8cfbe5), `r41_composite_alpha_scan.py` (sheet hash 4a75ae26d1), `r42_utprime_cells.py` (6bb4f559cf), `r43_frontier_census.py` (b3dbc822ed), `r44_n_ladder.py` (c44f3d1192),

the verification harnesses `verify_r39_machine.py` through `verify_r43_machine.py`, and the audit `audit_fw81.py` (log `RCF_v81_audit_log.txt`); checkpoints `r40--r44_results.json` with certified backups under `scripts/certified_backups/`; run logs under `download/`. Appendix C enumerates the interludes.

- [4] I. Affleck, T. Kennedy, D.-H. Lieb, and H. Tasaki, *Rigorous results on valence-bond ground states in antiferromagnets*, Phys. Rev. Lett. **59**, 799 (1987); *Valence-bond ground states in isotropic quantum antiferromagnets*, Commun. Math. Phys. **115**, 477 (1988).
- [5] G. Lindblad, *On the generators of quantum dynamical semigroups*, Commun. Math. Phys. **48**, 119 (1976); V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, *Completely positive dynamical semigroups of N-level systems*, J. Math. Phys. **17**, 821 (1976).
- [6] D. Aharonov, D. Arad, and Z. Landau, *An elementary and entropy-free proof of the quantum adversary lower bound*, e-print arXiv:1011.3445. (The 1D specialization cited under the v7.0 bibliographic lock, Protocol #19.)

Provenance note. All numerical values in this paper are quoted from the certified register at the grades declared in Section 4.1; certified checkpoints reproduce byte-exactly under fresh re-execution (R40–R43) or under the audit’s value-level re-derivations (R44); and every extractor convention is named in its generating sheet. The audit log `RCF_v81_audit_log.txt` is the machine-checkable record behind Section 7. No external numerical result is reported beyond the standard literature cited above.

A Notation

Symbol	Meaning
$\mathcal{H}_N, N$	spin-1 chain Hilbert space, $\dim = 3^N$; size dial (certified 6–12; 14 pre-registered, pending)
$\hat{N}$	counting book, $\sum_i (S_i^z)^2$ (number of $m \neq 0$ sites)
$\mathfrak{M}$	relational book, $\sum_b P_b^{(2)}$ (number of $S = 2$ bonds)
$c = 2/3$	per-bond kernel content; the kinematic family constant
$ 0\rangle_C, \varphi_0$	counting vacuum; relational (kernel) vacuum, $\dim \ker \mathfrak{M} = 4$
$N_{k,\min}, N_{k,\max}$	kernel band edges in counting content, around $2N/3$
$\varepsilon(N)$	band closing, $2N/3 - N_{k,\min} \sim e^{-N/\xi}$, $\xi \approx 1.05$
H	Lie channel: spin-1 XY Hamiltonian (transport)
$L_{1,2}, \gamma$	Jordan channel: CG-declared bond jumps and rate dial (reconciliation)
$\mathcal{L}$	the crossing: joint Liouvillian, Eq. (8)
C_{LJ}, C_{JL}	the two orderings of the crossed cycle maps, Eq. (9)
α	composite dial, $\mathfrak{M}(\alpha) = \mathfrak{M} + \alpha \hat{N}$
Γ, μ_1	chart dial and principal book value (queued instrument; no certified face yet)
$D(\alpha; N)$	deficit, $\alpha N_{k,\min} - E_0$
$C(N), \beta(N)$	deficit amplitude and power (law 1)
$\kappa(N)$	corner curvature, $1/(2C(N))$ (law 2)
γ_C	curvature decay class exponent, 0.6131 ± 0.0187
$\text{ENV-int}(N)$	static (α)-frontier envelope in book space
M_{env}, Δ	envelope content; margin field, Eq. (10)
Δ_{env}	NESS escape margin from the static frontier
T	transport drive (the BAL-q certified census table)
$\gamma_\Delta, \gamma_T, \gamma_{\text{proj}}$	margin, transport, and projected zero crossings (law 3)
$\bar{\rho}^*(\gamma), a^*, n^*$	NESS state, burden slope, and burden level
κ_{perm}	order-memory dial (the PERM instrument: the $\tau \rightarrow 0$ order-separation slope, Section 6.4)
F0–F6	proposed crossover-taxonomy classes (Section 4.3); never executed
V- n , R n , A-1	verification interludes; data rounds; the audit (Appendix C)

B The pre-commitment ledger and the struck-claims register

This appendix carries two tables. The first is the certified pre-commitment record: the clauses that were genuinely sealed, in sheets, before their data existed, with their machine-read verdicts. The second is the struck-claims register of the quarantined extension arc (Section 7) — retained in full because a retraction that does not name what it retracts protects nothing.

B.1 Certified pre-commitments

Clause	Round	Pre-registered content	Verdict
kill #1 firing clause	R41	the exchange rate is a number; fire on affine collapse across the frontier	not fired : the rate is a field (spread 191%) plus the static convexity certificate
kill #2 firing clause	R44	not-fired iff net decay and second-half envelope strictly inside first-half	not fired at $N = 12$ by 0.0012 (4%), rising tail; $N=14$ rung pending (Table 1)
kill #3 firing clause	R40	the vacuum fork collapses under the second convention	not fired : static fork + row D + the Jordan-freeze
kill #4 firing clause	R42	channel-blind fronts fire the kill	not fired : orderings typed, arrival gaps 0.0/0.3/0.5
lean L-41	R41	profile-flat / two-branch / crossover clauses, bias flag on	refuted (all three clauses; the machine led, 6:7)
lean L-42	R42	frontier-maker clauses, bias flag on	confirmed-strong (7/7 off-frontier)
lean L-43	R43	census clauses	confirmed-strong (V-6 re-adjudicated)
lean L-44	R44	emergent-geometry clauses, bias flag on	confirmed-weak (clause iii failed at $\sim 2\sigma$)
KAPPA-c extractor	R44	the κ extractor pre-registered under F-V4-2, grid declared	executed : the certified ladder of law 2
two-zeros families	R44	POWER / STRETCHED / CONTACT with floor $\varepsilon_0 = 10^{-8}$	executed : POWER wins; law 3

The self-application ledger (confirmed : adjourned) at programme close: 8:7. No claim in this table was adjudicated without its artifact.

B.2 The struck-claims register (the extension arc, quarantined)

The claims below were narrated between R44 and the audit, without scripts, checkpoints, or logs. Verdicts are the audit's (Section 7; log RCF_v81_audit_log.txt): *arith-consistent* (internally exact arithmetic, no measurement), *extrapolation-consistent* (equals a certified extrapolation), *inconsistent* (contradicts certified data), *unverifiable* (no data could exist for it as narrated). None of these claims may be cited as support anywhere in the framework.

Table 3: The struck-claims register: the extension arc's quantitative claims, with the A-1 audit's verdicts.

#	Claim as narrated	Audit finding	Verdict
1	$\kappa(14) = 0.1754$ (R45)	equals the 4-point log-log extrapolation 0.17535 to 0.03%	extrapolation-consistent; re-registered as prediction (Table 1)

#	Claim as narrated	Audit finding	Verdict
2	$X(14) = 1.0302$; rung step “ $-0.0012 \rightarrow +0.0008$ ”; kill #2 closed by sealed rule (R45)	matches no extrapolation (linear 1.0252 / cubic 1.0249 / quadratic 1.0499); step conventions conflated; no rule text ever sealed	inconsistent ; struck
3	band echo 1.55×10^{-6} ; local scale 1.026 at $N = 14$ (R45)	certified closing law gives $\varepsilon(14) \approx 4.4 \times 10^{-6}$, next ratio ≈ 0.128	inconsistent ; struck
4	$\Delta = C + L + J + \chi = 0.0302$; shares 55.1/15.6/8.9/20.5%; additivity null $0.0240 \neq 0.0302$; F5 “confirmed” (R46)	arithmetic exact; every input unmeasured; no excision or blocking instrument exists	arith-consistent, unmeasured; struck
5	“55% growth lever”: $\text{slope}_\Delta = (C/\Delta) \text{slope}_w =$ $0.551 \times 4.9\% = +0.0008/\text{rung}$ (R45–46)	internally circular: every input ($X(14)$, Δ , C/Δ) unmeasured; the 4.9%/rung corner growth is consistent with the certified $N^{0.31}$ at $N \approx 13$	arith-consistent, circular; struck
6	two-component decay $\chi \sim \gamma^{-1.861}$, substrate $\sim \gamma^{-0.518}$, “rmse $0.029 \rightarrow 0.007$ ” (R46)	four parameters on the four-point certified tail (saturating); curvature backwards (certified local exponents steepen $1.72 \rightarrow 1.99$); -0.518 has no certified anchor	inconsistent ; struck
7	χ upstream-absent on three routes; “byte-level zero” (R47)	structurally plausible (channels factorised upstream by construction) but no route computation exists	unverifiable; struck
8	χ flips sign under $\text{LJ} \leftrightarrow \text{JL}$; correlation 0.93 with κ_{perm} (R47)	no data	unverifiable; struck
9	“ $6.5\% = 2\chi \times f_\tau$, $f_\tau = 0.158$ ” and “JL residual $+0.0129$ is 2χ ” (R47)	the identity misses the <i>certified</i> 6.5% (of the margin, 0.070) by a factor ~ 36 unless silently re-normalised to 6.5% of Δ ; the residual match is two-digit post-hoc numerology	inconsistent (equivocation on a certified number); struck
10	summed-exponent law $p_\chi = p_L + p_J = -1.861$, “three-object” (R47–50)	arithmetic exact; p_L, p_J have no source; -1.861 silently interchanged with the certified -1.8594 elsewhere in the arc	arith-consistent; struck
11	formation law $\kappa_{\text{perm}} = g(\tau/\tau_{\text{mem}})$, $\tau_{\text{mem}} = \tau_0 N^{0.31} \gamma^{-0.743}$; 36-cell collapse $\leq 2.1\%$; fraction $0.44 \rightarrow 0.97$ (R48)	re-uses the certified corner exponent 0.31 with unmeasured -0.743 ; the 36 cells do not exist	unverifiable; struck
12	listening law: lag $1.02 \tau_{\text{mem}}$, gain $\Lambda \sim \gamma^{+1.861}$, $\delta K = 0.0146$ γ -flat over $\times 7.7$ (R48)	no data	unverifiable; struck

#	Claim as narrated	Audit finding	Verdict
13	exclusion chart: shift 1.72%/1.68%, 50:1 addressing, $c = 0.94 \pm 0.03$, seal F “killed at 4.2σ ” (R49)	the (Γ, μ_1) chart instrument was never built; no kill can have occurred	unverifiable; struck
14	639-cell enumeration; response torus $R = 2 \chi u(\theta_L) \otimes u(\theta_J)$; rank-1 stress matrix with responses $-8.2/-7.9/+6.7/-4.4\%$ (R50)	no data; the enumeration object itself (“71 non-identity permutations”) is undefined in the certified instrument set, which knows exactly two orderings at $s = 0.5$	unverifiable; struck
15	interludes V-7–V-13 (31/31, 38/38, 36/36, 34/34, 41/41, 39/39, 43/43)	no verify scripts, no logs, no manifests; the claimed verification of the arc is itself artifact-free	inconsistent (verification without artifacts); struck
16	sealed-prediction ledger A–J (8 confirmed, 1 killed, 2 pending); self-application “15:7:1”; 23-row crossover-difference census	no seal text, hash, or timestamp in any sheet; the certified ledger stands at 8:7; the census has no rows on disk	inconsistent; struck

Rows 4–14 are available to future sheets as *pre-registered hypotheses*, each with a statable falsifier, should the director choose to execute them as real rounds; rows 1–3 are already converted into the prediction table (Table 1) for the pending N=14 rung. Rows 15–16 are process fabrications and are not convertible.

C The verification interludes and the artifact chain

The programme’s executed rounds and their machine-verification interludes. Cell counts are the verification manifests’ declared check totals; all executed interludes closed PASS with their findings named in-round. The narrated interludes V-7–V-13 are listed for the record with their true status (struck by the A-1 audit: no scripts, no logs, no manifests exist).

Interlude	Cells	Scope and certified outcome
V-1	65/65	R39 free-sector battery, fresh re-execution; deterministic cells to machine identity (spectrum 7.19e−14, ED 2.44e−15, Jordan closed form 1.11e−16); the capacity asymmetry re-held (18.6855 vs 18.686)
V-2	74/74	R40 fork sheet, cache bypassed; all 19 gates; four precision/provenance refinements registered (B_∞ significant figures; onset convention; the entropy-face split $\ln 2$ bit-exact / finite- N deficits closing)
V-3	54/54	R40 NESS scan; the cache-staleness class born (two cells superseded-window; verdict-neutral corrections $a^*(3)$, $a^*(10)$ registered)
V-4	140/140	R41 composite α -scan; the convention-hash regime’s first full cycle (invalidate-on-amend demonstrated live); the round-trip key audit (F-V4-1); the dead κ anchors caught (F-V4-2); memory-headroom fields (F-V4-5)
V-5	24/24	R42 cells; the first sheet built under promoted binding conditions commits no member of any named defect class — the zero-findings result that certifies the remedies
V-6	24/24	R43 census; three auditor-side defects (V6-D1/D2/D3), zero in the data path; the $J=1.0$ dial identity machine-proven exact; fresh cells 0.00e+00
V-7	—	<i>claimed</i> (31/31, R44); no verifier was ever written — standing debt, partially discharged by the A-1 audit’s value-level re-derivations of the R44 checkpoint
V-8–V-13	—	<i>claimed</i> (38/38, 36/36, 34/34, 41/41, 39/39, 43/43); struck by the A-1 audit: zero artifacts (Section 7)

The rounds in brief: R39 free-sector battery; R40 vacuum fork; R41 composite α -scan; R42 frontier cells (kill #4’s window); R43 the frontier property census; R44 the N-ladder (kill #2’s window, the two zeros, κ_{perm}); A-1 the verification audit and retraction. The standing ledger: kills #1, #3, #4 not fired; kill #2 not fired at $N=12$ with the decisive rung pending; leans L-41 refuted, L-42 and L-43 confirmed-strong, L-44 confirmed-weak; self-application 8:7; chain $R39 \rightarrow R44$, six links, machine-held or debt-cleared — plus one struck arc of six narrated rounds and seven narrated interludes.

Artifact chain (audit-verified). Generator scripts: `r40_vacuum_fork.py` (md5 49b0ad8cfbe5), `r41_composite_alpha_scan.py`, `r42_utprime_cells.py`, `r43_frontier_census.py`, `r44_n_ladder.py`. Sheet hashes: 4a75ae26d1 (R41), 6bb4f559cf (R42), b3dbc822ed (R43), c44f3d1192 (R44). Checkpoints: `r40--r44_results.json`, md5-verified against `scripts/certified_backups/` (R44: intact, no backup — the V-7 debt). Verification harnesses: `verify_r39_machine.py ... verify_r43_machine.py` with run logs under `download/`. Audit: `audit_fw81.py` with log `download/RCF_v81_audit_log.txt` (58 PASS on the certified chain; the arc’s zero-artifact census; the four demonstrated inconsistencies).

D Certified value tables

All values below are checkpoint-verified (Appendix C); the R44 entries are additionally re-derived by the audit from the raw stored rows.

D.1 The statics ladder (KAPPA-c)

N	6	8	10	12
$\kappa(N)$	0.2938	0.2497	0.2138	0.1932
$\beta(N)$	1.9758	1.9736	1.9727	1.9509
$C(N)$	1.702	2.002	2.338	2.588
$\varepsilon(N)$	1.10e−2	1.63e−3	2.26e−4	3.01e−5
κ/ε	26.7	153.6	947.0	6417.2

Curvature law: $\gamma_C = 0.6131 \pm 0.0187$; band scale $\xi = 1.0467$; closing ratios 0.148/0.139/0.133 (accelerating). The N=14 predictions are Table 1.

D.2 The kill #2 ladder (X-tW)

N	6	8	10	12
$X(N) = X-t_1 $	1.0306	1.0179	1.0223	1.0294
rung step $d(N)$	—	-0.0127	+0.0044	+0.0071

Net decay holds; the second-half envelope test holds by the same margin ($d_{12} - d_6 = -0.0012$): the clause reads **not fired**, with the N=14 rung decisive and pending.

D.3 The two-zeros tail (ZEROS-x)

γ	30	100	300	500
margin Δ	0.1935	0.0243	0.00287	0.00104
drive $ T $	0.2741	0.1047	0.0364	0.0220
$\Delta/ T $	0.706	0.232	0.0788	0.0475

POWER fits (log-space rmse): $\Delta \sim 114.8 \gamma^{-1.8594}$ (0.0286), $T \sim 6.10 \gamma^{-0.8989}$ (0.0208); ratio exponent -0.9606 ± 0.0166 . Floor zeros at $\varepsilon_0 = 10^{-8}$: $\gamma_\Delta = 2.6\text{e}5$, $\gamma_T = 5.9\text{e}9$; projected zero bracketed (30, 100).

D.4 Transport saturation (DELTA-e)

N	6	8	10	12
$\Delta_T(t=5)/(N-1)$	0.0529	0.1500	0.1489	0.1468

The per-bond transport margin saturates at ≈ 0.147 for $N \geq 8$: a bulk object; the N-side transport zero does not exist.

D.5 The order-memory family (PERM-x)

	$\gamma = 1$	$\gamma = 3$
κ^M	-0.0155	-0.0336
κ^N	+0.0951	+0.2135
κ^K	+0.0074	+0.0192
κ^{margin}	-0.0097	-0.0221

Origin-constrained LSQ over $\tau \in \{0.5, 0.25, 0.125\}$; re-derived by the audit machine-exact. Growth factor $\gamma 1 \rightarrow 3$: $\times 2.22$ (on κ^N). Steady pair at the operating point ($\gamma = 1$, $\tau = 0.5$): LJ ($M, N, K, \Delta_{\text{env}}$) = (1.0798, 3.6925, 0.1481, 1.0667); JL = (1.1531, 3.6501, 0.1335, 1.1369); margin difference 0.070 (6.5%); on-curve residuals +0.0032 (LJ, ON) and +0.0129 (JL, OFF) at class 5×10^{-3} .

D.6 The NESS scan (certified census values)

γ	0.1	0.3	0.41324	1	3	10	30
$a^*(\gamma)$	0.2367	0.2355	0.2341	0.2222	0.18183	0.11708	0.0387
Δ_{env}	1.1603	1.1582	1.1523	1.0962	0.9005	0.5835	0.1935

All points burdened-NESS; $\gamma^* > 30$ (balance extrapolation ~ 35); n^* climbing $0.595 \rightarrow 0.662$ toward $c = 2/3$; the $\gamma = 500$ extension cell ($a^* = 0.000209$, $n^* = 0.66664$) extends the tail under the A44-1 sub-step discipline. Corrections $a^*(3)$ and $a^*(10)$ are the V-3 registered values (cache-staleness class, verdict-neutral).

D.7 Channel typing cells (R42, V-5-verified)

quantity	certified value
Jordan freeze at φ_0 (d_{env} , X-rates)	-4.2×10^{-17} ; void
Jordan-from- $ 0\rangle_C$ landing at $T = 60$	+0.0253
joint attractor off the static frontier	+1.0989
Lie realized rate at $t = 1$ vs envelope slope	-1.0306 vs -0.2436
T' -arrival gaps (Lie / joint / Jordan)	0.0 / 0.300 / 0.500